

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. Examination, 2024
CHE – 1104, Chemistry

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) Describe Bohr's atomic model. Mention its limitations. 5
 b) What are quantum numbers? Calculate the maximum number of electrons present in 3rd energy level of an atom using four quantum numbers. 1+3
 c) What is meant by Pauli's exclusion principle? Explain with example. 3
 d) Write the electronic configuration of the following atoms or ions- Cr, Fe²⁺, Br, Cu 2

2. a) What do you mean by hydrogen bonds? 2
 b) Why are the shapes of NH₃ and H₂O different from CH₄? 3
 c) What is Co-ordination bond? How is it formed? Explain with example. 4
 d) What is meant by ionic bond? Explain – the reason of ionic compounds of covalent character. 5

3. a) Define ionization potential. Why the first ionization of N atom is greater than O atom? 1+2
 b) Why the Electron affinity of Cl is more than F? 2
 c) Discuss the trends of atomic size across a period and down a group of the periodic table. 3
 d) What are colligative properties? Explain. 2
 e) What is Gibb's phase rule? Describe the phase diagram of mono component system like H₂O. 1+3

4. a) What do you understand by activated complex, and activation energy of a chemical reaction? 3
 b) What is half-life period of a reaction? Differentiate between 'order' and 'molecularity' of a reaction. 4
 c) What is the rate constant? Derive an expression for the rate constant of a first order reaction. 4
 d) 50% of a first order reaction is complete in 23 minutes. Calculate the time required to complete 90% of the reaction. 3

5. a) What is buffer solution? Write the mechanism of acid buffer action. 1+3
 b) What is an indicator? How an indicator changes its color during titration? 1+2
 c) What do you understand by ionic product of water? 1
 d) Calculate the pH value of 0.1M H₂SO₄ 2
 e) What do you mean by oxidizing and reducing agent? Balance the equation: FeSO₄ + K₂Cr₂O₇ + H₂SO₄ → ? 1.5+2.5

6. a) What is meant by chemical equilibrium? Why chemical equilibrium is called dynamic equilibrium? 4
 b) State and explain the law of mass action. Deduce the mathematical formulation of the law of mass action. 4
 c) Discuss the application of Le-Chatelier principal to the following:

$$2NH_3 \rightleftharpoons N_2 + 3H_2$$
 3
 d) At 25°C temperature and 1 atm pressure PCl₅ is 80% dissociated into PCl₃ and Cl₂ respectively. Calculate the partial pressure of PCl₃ and Cl₂ and K_p and K_c. 3

7. a) Define standard electrode potential and state the difference between Electrochemical cell and Electrolytic cell. 3
 b) Explain the EMF of the cell: Zn | Zn²⁺(0.001M) || Ag⁺(0.1M) | Ag 3
 c) Discuss the enthalpy of a system. 2
 d) Derive the equation for work done of an ideal gas in isothermal reversible expansion. 6